Solution for module 8 exercises

CH 11, #8, 11, 19, 24, 28

#8 - a, b, c - see Excel worksheet.

d. The model in part c provides a better fit.

11. $\hat{y} = 17.6 + 3.8x_1 - 2.3x_2 + 7.6x_3 + 2.7x_4$

$n = 30 \quad SST = 1805 \quad SSR = 1760$

a) $\alpha = .05$

$SSE = SST - SSR = 45$

$MSR = \frac{SSR}{4} = 440$

$MSE = \frac{SSE}{n-p-1} = \frac{45}{30-4-1} = 1.8$

$F = \frac{MSR}{MSE} = \frac{440}{1.8} = 244$ $\quad df = \frac{p}{n-p-1} = \frac{4}{25}$

p-value ≈ 0

p-val $\leq \alpha$ ∴ The overall relationship <u>is</u> significant.

b) $SSE(x_1 \ldots x_4)$

Obtained in part a.

$SSE = 45$

c) $SSE(x_2, x_3)$

① SST & SSR for x_1, x_2, x_3, x_4 SST 1805 SSR 1760

② SST & SSR for x_2, x_3 = SST 1805 SSR 1705

$SSE = \underset{③}{SST} - \underset{②}{SSR} = 100$

d) Use equation 11.15 to find the f value for the difference.

$$\frac{\frac{100-45}{2}}{1.8} = 15.28$$

$df = \frac{2}{27}$

$p \approx 0$

x_1 and x_4 Contribute significantly to the model.

OPTION 1

#19 Generate "Best Subsets Regression" with minitab,

OR - Try all combinations of regression models.

Best R^2 = Age + Pressure + Smoker.

$$\text{Risk} = -91.76 + 1.08(\text{age}) + .25(\text{pressure}) + 8.74(\text{smoker})$$

24. $a = 0.05$

Steps: 1) Generate dummy value matrix
2) Represent real data using dummy values. y = actual value.
3) check F or p value for regression model.

Hypothesis test = H_0 $\beta_1 = \beta_2 = \beta_3 \ldots = 0$

$a = 0.05$
$p = 0.09$

$p > a \therefore$ Fail to reject H_0

The paint drying times are similar.

28. Durbin-Watson Test

See Excel example.

You need to do a regression to get $\hat{y}$ (intercept + slopes).

Calculate errors + etc...
Obtain $d =$ value.

$d = 1.6$ Lookup table $n = 25$
$\alpha = 0.05$
$k = 4$

$d_u = 1.77$
$d_L = 1.04$ $\leftarrow$ $\boxed{\begin{matrix} d \\ 1.6 \end{matrix}}$.

$d_L \leq d \leq d_u$ – The test is inconclusive.

Does y correlate to previous values of y over time?